

5-Day SDI Lab Automation FDP

Workshop Agenda — Version 2.0

Mahaveer. Rajendra. Savanur | Head of Engineering & AI Systems

DAY 1 — Foundations Part 1: "From Manual to Automated"

Time	Session
8:00 – 9:00	Introduction & Workshop Overview
9:15 – 10:30	Environment Setup & Verification (config.py)
10:45 – 12:30	Oscilloscope & Waveform Generator — demos + Exercises 1.1, 1.2
1:30 – 3:30	Practice, Bonus Challenges & Office Hours
3:45 – 5:00	Code Review, Q&A, Day 1 Wrap-Up

DAY 2 — Foundations Part 2: AWG, Logic Analyzer & MiM

Time	Session
8:00 – 10:30	Day 1 Recap • AWG (Ex 2.1) • Logic Analyzer (Ex 2.2)
10:45 – 12:30	Day 1 Capstone: Frequency Response Tester
1:30 – 2:30	Capstone Presentations
2:45 – 5:00	Multi-Instrument Mode Fundamentals + Ex 2.3

DAY 3 — Integration & Advanced Techniques

Time	Session
8:00 – 10:30	MiM Hands-On: Frequency Sweep, AM Modulation, Filter Characterization
10:45 – 12:15	Real-Time Streaming (Ex 3.1)
1:15 – 3:15	FFT, SNR & THD Analysis (Ex 3.2)
3:30 – 5:00	Day 2 Capstone Development

DAY 4 — Real-World Applications & Final Projects

Time	Session
8:00 – 9:15	Journey Review + Day 2 Capstone Presentations
9:30 – 11:00	Industry Case Studies + 5 Automation Patterns
11:15 – 12:30	Pattern Matching + Final Capstone Planning
1:30 – 3:30	Final Capstone Implementation (Options A/B/C)
3:45 – 5:00	Final Presentations, Awards, Day 4 Close

DAY 5 — AI-Augmented Curriculum Design for SDI Labs

Time	Session
8:00 – 8:30	Welcome & Day 5 Overview
8:30 – 10:30	Moku Custom Instrument (MCC) Session — threshold detector hands-on
10:45 – 12:30	AI-Assisted Curriculum Mapping (Claude + GPT, free tiers)
1:30 – 4:30	AI-Assisted Experiment Design (GPT ideation → Claude design → faculty verification)
4:30 – 5:00	Sharing, Wrap-Up & Certificate Distribution

Daily breaks: 15 min mid-morning and mid-afternoon; lunch 60 min. Total programme: 40 hours across 5 days.